Creating a Simple MAVEN TO GRADLE Project

Step 1: Create a new maven project in Devops by using the command

mvn archetype:generate -DgroupId=com.example -DartifactId=MyMavenToGradle -DarchetypeArtifactId=maven-archetype-quickstart -DinteractiveMode=false

```
vboxuser@niranjan: ~/devops
vboxuser@niranjan:~$ cd devops
vboxuser@niranjan:~/devops$ mvn archetype:generate -DgroupId=com.example -DartifactId=MyMaventogradle -DarchetypeArtifactId=maven-archetype-w
ebapp -DinteractiveMode=false
[INFO] Scanning for projects...
[INFO]
[INFO] ------------------< org.apache.maven:standalone-pom >-------------------
[INFO] Building Maven Stub Project (No POM) 1
[INFO] --------------------------------[ pom ]---------------------------------
[INFO]
[INFO] >>> maven-archetype-plugin:3.4.1:generate (default-cli) > generate-sources @ standalone-pom >>>
[INFO]
[INFO] <<< maven-archetype-plugin:3.4.1:generate (default-cli) < generate-sources @ standalone-pom <<<
[INFO]
[INFO]
[INFO] --- maven-archetype-plugin:3.4.1:generate (default-cli) @ standalone-pom ---
[INFO] Generating project in Batch mode
[INFO] ----------------------------------------------------------------------------
[INFO] Using following parameters for creating project from Old (1.x) Archetype: maven-archetype-webapp:1.0
[INFO] ----------------------------------------------------------------------------
[INFO] Parameter: basedir, Value: /home/vboxuser/devops
[INFO] Parameter: package, Value: com.example
[INFO] Parameter: groupId, Value: com.example
[INFO] Parameter: artifactId, Value: MyMaventogradle
[INFO] Parameter: packageName, Value: com.example
[INFO] Parameter: version, Value: 1.0-SNAPSHOT
[INFO] project created from Old (1.x) Archetype in dir: /home/vboxuser/devops/MyMaventogradle
[INFO] ------------------------------------------------------------------------
[INFO] BUILD SUCCESS
[INFO] ------------------------------------------------------------------------
[INFO] Total time:  8.907 s
[INFO] Finished at: 2026-04-15T23:31:32+05:30
[INFO] ------------------------------------------------------------------------
vboxuser@niranjan:~/devops$
```

Step 2: Edit pom.xml

```
<project xmlns="http://maven.apache.org/POM/4.0.0"
xmlns:xsi="http://www.w3.org/2001/XMLSchema-instance"
xsi:schemaLocation="http://maven.apache.org/POM/4.0.0
http://maven.apache.org/maven-v4_0_0.xsd">
<modelVersion>4.0.0</modelVersion>
<groupId>com.example1</groupId>
<artifactId>MyMaventoGradle</artifactId>
<packaging>jar</packaging>
<version>1.0-SNAPSHOT</version>
<name>MyMaventoGradle</name>
<url>http://maven.apache.org</url>
<!--  Properties: Customize Java version or plugin versions  -->
<properties>
<maven.compiler.source>11</maven.compiler.source>
<maven.compiler.target>11</maven.compiler.target>
```

```
</properties>
<dependencies>
<dependency>
<groupId>junit</groupId>
<artifactId>junit</artifactId>
<version>3.8.1</version>
<scope>test</scope>
</dependency>
</dependencies>
<!--  Build: Configuring plugins and build settings  -->
<build>
<plugins>
<!--  Example: Maven Compiler Plugin to compile Java code  -->
<plugin>
<groupId>org.apache.maven.plugins</groupId>
<artifactId>maven-compiler-plugin</artifactId>
<version>3.8.1</version>
<configuration>
<source>1.6</source>
<target>1.6</target>
</configuration>
</plugin>
<!--  Example: Maven Surefire Plugin to run tests  -->
<plugin>
<groupId>org.apache.maven.plugins</groupId>
<artifactId>maven-surefire-plugin</artifactId>
<version>2.22.2</version>
</plugin>
<plugin>
<groupId>org.apache.maven.plugins</groupId>
<artifactId>maven-jar-plugin</artifactId>
<version>3.1.0</version>
<configuration>
<archive>
<manifestEntries>
<Main-Class>com.example1.App</Main-Class>
</manifestEntries>
</archive>
</configuration>
</plugin>
</plugins>
</build>
</project>
```

Step 3: Execute the Tree command to display the Tree

```
es   Terminal                                                    Apr 15 23:35
vboxuser@niranjan: ~/devops/MyMaventogradle
vboxuser@niranjan:~/devops$ cd MyMaventogradle/
vboxuser@niranjan:~/devops/MyMaventogradle$ ls
pom.xml  src
vboxuser@niranjan:~/devops/MyMaventogradle$ tree
.
├── pom.xml
└── src
    └── main
        ├── resources
        └── webapp
            ├── index.jsp
            └── WEB-INF
                └── web.xml

5 directories, 3 files
vboxuser@niranjan:~/devops/MyMaventogradle$
```

Step 4: Execute the command mvn clean install

```
vboxuser@niranjan: ~/devops/MyMaventogradle
vboxuser@niranjan:~/devops/MyMaventogradle$ mvn clean install
[INFO] Scanning for projects...
[INFO]
[INFO] --------------------< com.example:MyMaventoGradle >----------------------
[INFO] Building MyMaventoGradle 1.0-SNAPSHOT
[INFO] --------------------------------[ jar ]---------------------------------
[INFO]
[INFO] --- maven-clean-plugin:2.5:clean (default-clean) @ MyMaventoGradle ---
[INFO]
[INFO] --- maven-resources-plugin:2.6:resources (default-resources) @ MyMaventoGradle ---
[WARNING] Using platform encoding (UTF-8 actually) to copy filtered resources, i.e. build is platform dependent!
[INFO] Copying 0 resource
[INFO]
[INFO] --- maven-compiler-plugin:3.8.1:compile (default-compile) @ MyMaventoGradle ---
[INFO] No sources to compile
[INFO]
[INFO] --- maven-resources-plugin:2.6:testResources (default-testResources) @ MyMaventoGradle ---
[WARNING] Using platform encoding (UTF-8 actually) to copy filtered resources, i.e. build is platform dependent!
[INFO] skip non existing resourceDirectory /home/vboxuser/devops/MyMaventogradle/src/test/resources
[INFO]
[INFO] --- maven-compiler-plugin:3.8.1:testCompile (default-testCompile) @ MyMaventoGradle ---
[INFO] No sources to compile
[INFO]
[INFO] --- maven-surefire-plugin:2.22.2:test (default-test) @ MyMaventoGradle ---
[INFO] No tests to run.
[INFO]
[INFO] --- maven-jar-plugin:3.1.0:jar (default-jar) @ MyMaventoGradle ---
[INFO] Building jar: /home/vboxuser/devops/MyMaventogradle/target/MyMaventoGradle-1.0-SNAPSHOT.jar
[INFO]
[INFO] --- maven-install-plugin:2.4:install (default-install) @ MyMaventoGradle ---
[INFO] Installing /home/vboxuser/devops/MyMaventogradle/target/MyMaventoGradle-1.0-SNAPSHOT.jar to /home/vboxuser/.m2/repository/com/example/
MyMaventoGradle/1.0-SNAPSHOT/MyMaventoGradle-1.0-SNAPSHOT.jar
[INFO] Installing /home/vboxuser/devops/MyMaventogradle/pom.xml to /home/vboxuser/.m2/repository/com/example/MyMaventoGradle/1.0-SNAPSHOT/MyM
[INFO] Installing /home/vboxuser/devops/MyMaventogradle/pom.xml to /home/vboxuser/.m2/repository/com/example/MyMaventoGradle/1.0-SNAPSHOT/MyM
aventoGradle-1.0-SNAPSHOT.pom
[INFO] ------------------------------------------------------------------------
[INFO] BUILD SUCCESS
[INFO] ------------------------------------------------------------------------
[INFO] Total time:  8.644 s
[INFO] Finished at: 2026-04-15T23:38:24+05:30
[INFO] ------------------------------------------------------------------------
vboxuser@niranjan:~/devops/MyMaventogradle$
```

Step 5: Now execute the command gradle init –type pom, we get build success

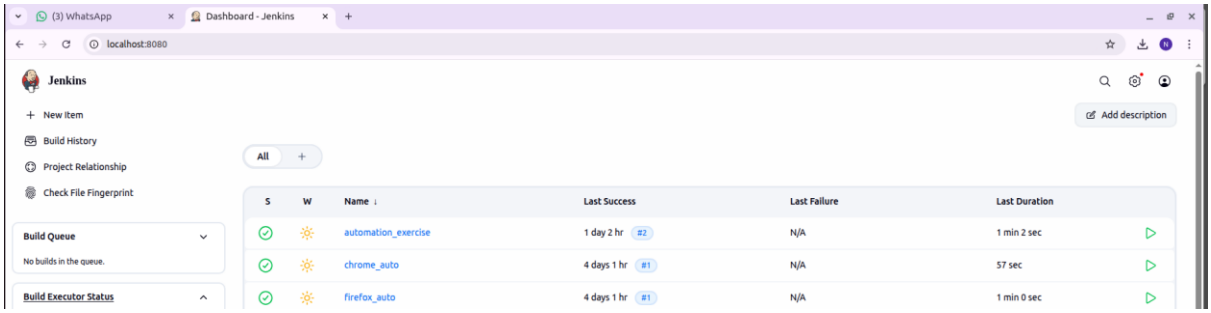
```
vboxuser@niranjan: ~/devops/MyMaventogradle
vboxuser@niranjan:~/devops/MyMaventogradle$ gradle init --type pom

> Task :init
Maven to Gradle conversion is an incubating feature.
<======-------> 50% EXECUTING [15s]
> :init
```

Step 6: Display the Tree Structure Generated using Tree Command

```
sumanth@sumanth-VirtualBox:~/Devops/MyMavenToGradle$ tree
.
├── build.gradle
├── gradle
│   └── wrapper
│       ├── gradle-wrapper.jar
│       └── gradle-wrapper.properties
├── gradlew
├── gradlew.bat
├── pom.xml
├── settings.gradle
├── src
│   ├── main
│   │   └── java
│   │       └── com
│   │           └── example
│   │               └── App.java
│   └── test
│       └── java
│           └── com
│               └── example
│                   └── AppTest.java
└── target
    ├── classes
    │   └── com
    │       └── example
    │           └── App.class
    ├── generated-sources
    │   └── annotations
    ├── generated-test-sources
    │   └── test-annotations
    ├── maven-archiver
    │   └── pom.properties
    ├── maven-status
    │   └── maven-compiler-plugin
    │       ├── compile
    │       │   └── default-compile
    │       │       ├── createdFiles.lst
    │       │       └── inputFiles.lst
    │       └── testCompile
    │           └── default-testCompile
    │               ├── createdFiles.lst
    │               └── inputFiles.lst
    ├── MyMavenToGradle-1.0-SNAPSHOT.jar
    ├── surefire-reports
    │   ├── com.example.AppTest.txt
    │   └── TEST-com.example.AppTest.xml
    └── test-classes
        └── com
            └── example
                └── AppTest.class

30 directories, 19 files
```

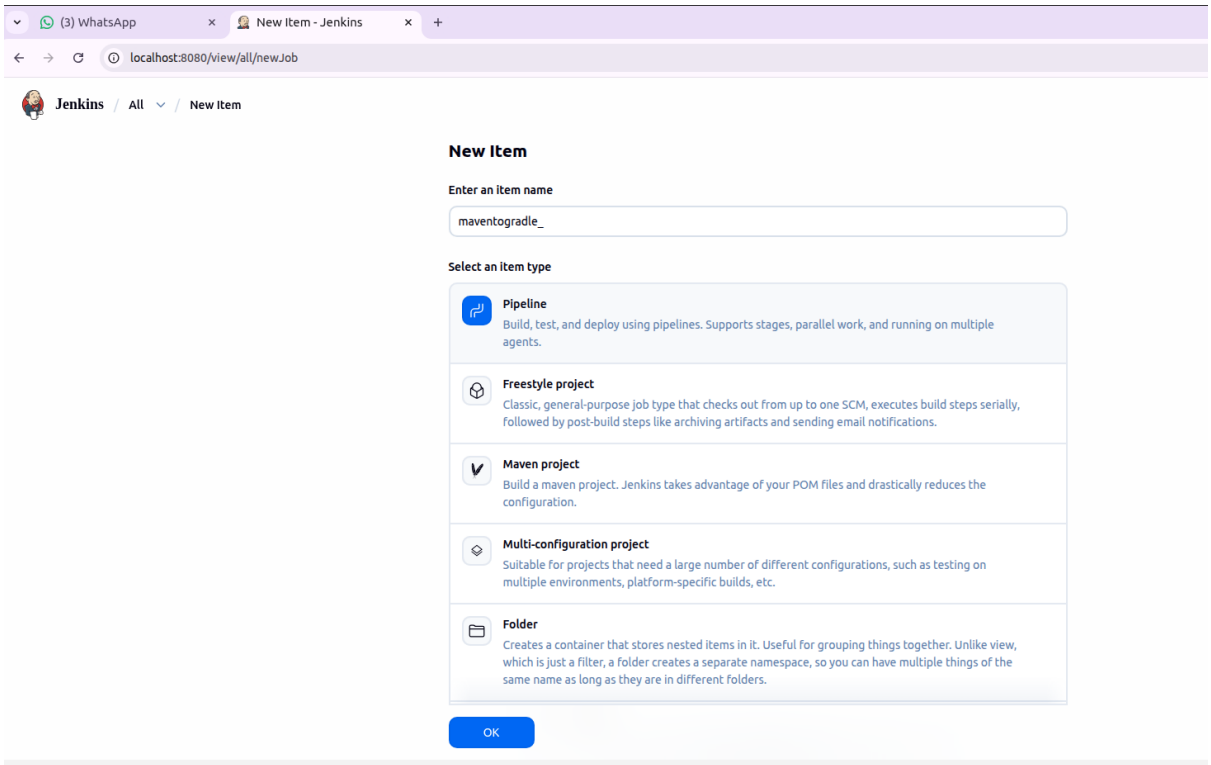
Jenkins Pipeline for MyMavenToGradle

Step 1: Open the Jenkins Dashboard to create a new item

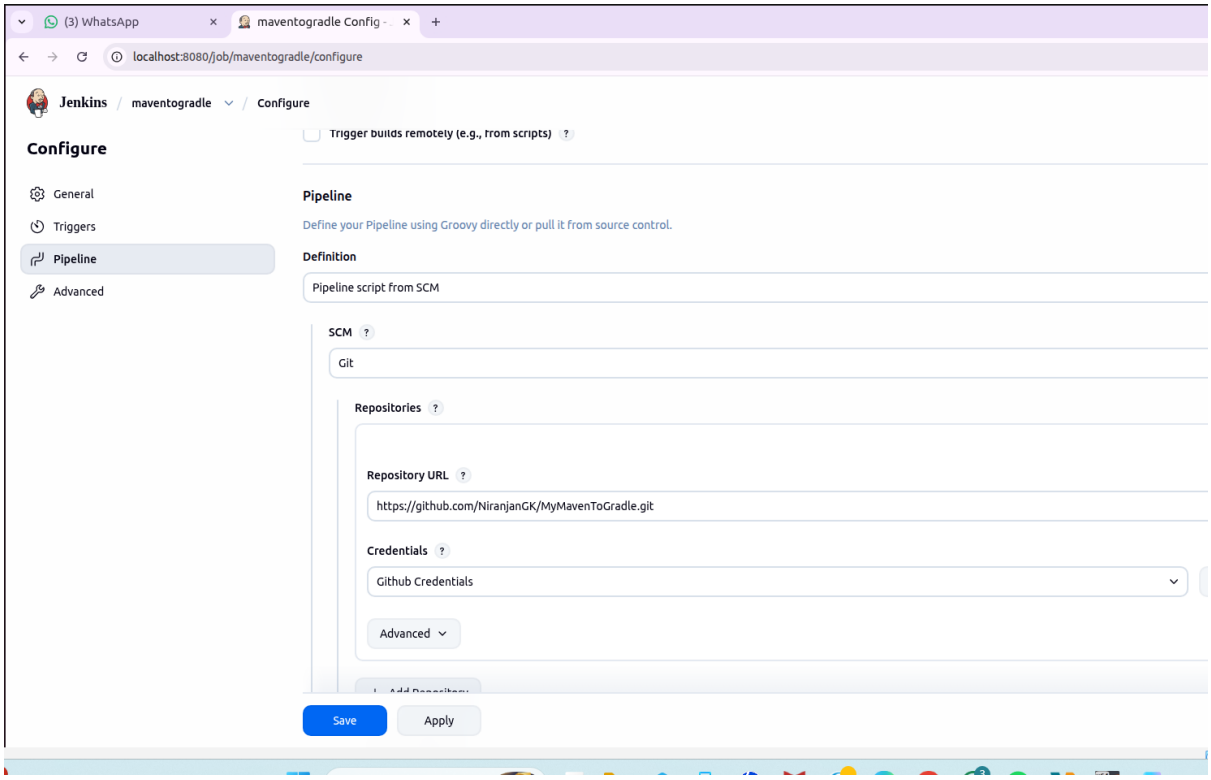


Step 2: Click on New Item in order for creating an New Item and thus give the name as "MyGradleApp" and Item type as Pipeline and

Create it



Step 3: Configure the settings by Defining the script as "Pipeline Script from SCM" and SCM as "Git". Thus, now as we have selected Git, we need to paste the Repository URL of the project given in GitHub and choose Credentials as "GitHub Credentials" which was created earlier.



<u>Step 4:</u> Now in Terminal, go to directory created for MyMavenAppSelenium05 and create an new file called "Jenkinsfile"

Step 5: Paste the available code inside the Jenkinsfile

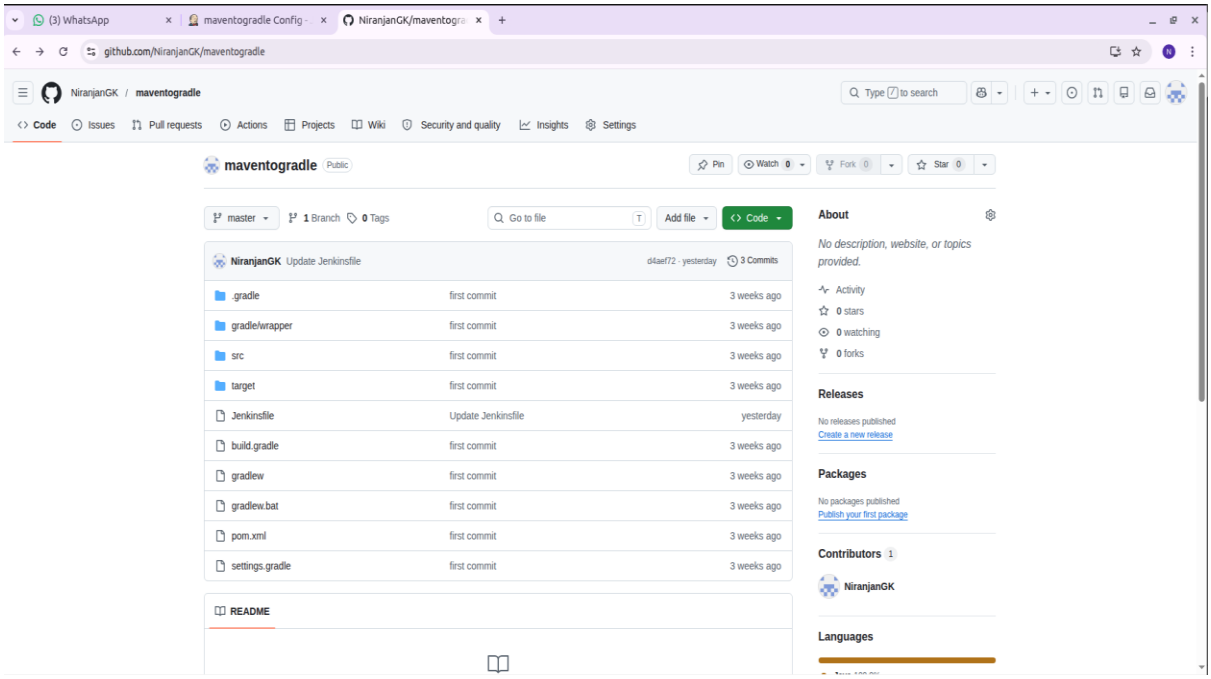
```
pipeline {
    agent any  // Use any available agent

    tools {
        gradle 'Gradle'  // Ensure this matches the name configured in Jenkins
        jdk 'JDK'
    }
    stages {
        stage('Checkout') {
            steps {
                git branch: 'master', url:
'https://github.com/NiranjanGK/maventogradle.git'
            }
        }

        stage('Build') {
            steps {
                sh 'gradle build'  // Run Gradle build
            }
        }

        stage('Test') {
            steps {
                sh 'gradle test'  // Run unit tests
            }
        }
```

```
        stage('Run Application') {
            steps {
                // Start the JAR application
                sh 'gradle run'
            }
        }



    }

    post {
        success {
            echo 'Build and deployment successful!'
        }
        failure {
            echo 'Build failed!'
        }
    }
}
```

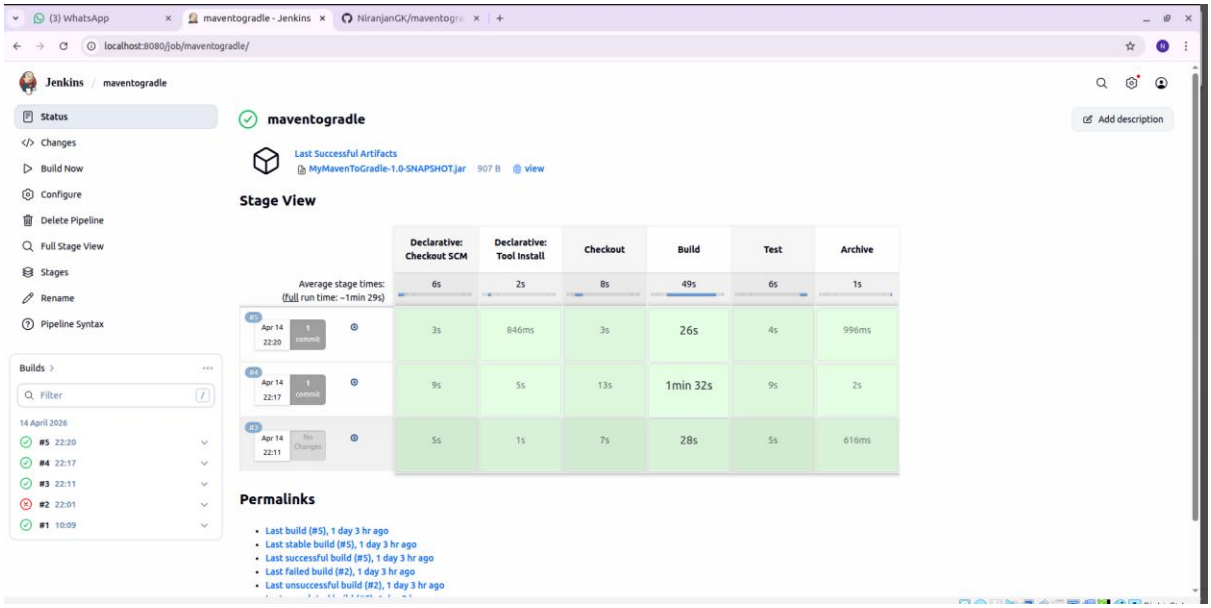
Step 6: We need to update pom.xml also in order to change the versions of source and target from 1.7 to 1.8

```
<!-- Example: Maven Compiler Plugin to compile Java code -->
<plugin>
<groupId>org.apache.maven.plugins</groupId>
<artifactId>maven-compiler-plugin</artifactId>
<version>3.8.1</version>
<configuration>
<source>1.8</source>
<target>1.8</target>
</configuration>
</plugin>
```

Step 7: Now we need to Push changes and commit by pushing the necessary "Jenkinsfile" to the respective Repo



Step 8: Now go to Jenkins and Build the Pipeline and thus check the Stage view for the successful execution.


(3) WhatsApp
maventogradle - Jenkins
NiranjanGK/maventogra
localhost:8080/job/maventogradle/
Jenkins
maventogradle
Status
Changes
Build Now
Configure
Delete Pipeline
Full Stage View
Stages
Rename
Pipeline Syntax
Builds
Filter
14 April 2026
#5 22:20
#4 22:17
#3 22:11
#2 22:01
#1 10:09
maventogradle
Add description
Last Successful Artifacts
MyMavenToGradle-1.0-SNAPSHOT.jar 907 B view
Stage View
Declarative: Checkout SCM
Declarative: Tool Install
Checkout
Build
Test
Archive
Average stage times:
(Full run time: ~1min 29s)
6s
2s
8s
49s
6s
1s
#5 Apr 14 22:20 1 commit
3s
846ms
3s
26s
4s
996ms
#4 Apr 14 22:17 1 commit
9s
5s
13s
1min 32s
9s
2s
#3 Apr 14 22:11 No Changes
5s
1s
7s
28s
5s
616ms
Permalinks
Last build (#5), 1 day 3 hr ago
Last stable build (#5), 1 day 3 hr ago
Last successful build (#5), 1 day 3 hr ago
Last failed build (#2), 1 day 3 hr ago
Last unsuccessful build (#2), 1 day 3 hr ago